

BiteBud

Competition roadmap, complete timeline & future prototype plan

A single operating plan for turning the current pressure-sensing bench system into a credible, competition-ready demonstration - while keeping safety, evidence, and realistic claims at the center.

THE NORTH STAR

Prove that a sensor-enabled teether concept can turn chewing patterns into simple, non-diagnostic insights for caregivers.

CURRENT POSITION

Dashboard and sensor firmware exist. Breadboard wiring is prepared. Hardware validation and calibration are the immediate next steps.

RECOMMENDED COMPETITION SLATE

1 / Conrad Challenge

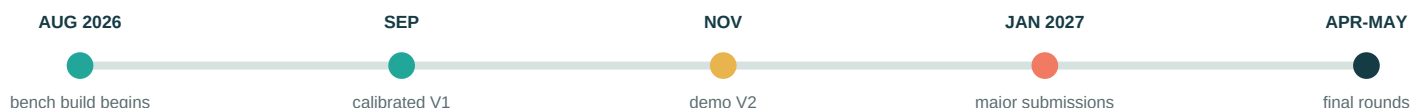
Flagship target: strongest fit for a working innovation, website, business case, and 3-5 minute demonstration.

2 / Diamond Challenge

Venture target: converts the same evidence into a 3-5 page narrative and 60-second introduction.

3 / Blue Ocean

Pitch target: reuses the business story in a strategy-led video due 21 February 2027.



Planning status. Conrad, Diamond, and Blue Ocean are the recommended 2026-27 slate. The local files do not confirm registration in any event. Confirm team eligibility and registration before treating the slate as committed.

Prepared 16 August 2026 | Team context: Jack Wang + Dhruv Kumar | Version 1.0

01 / FOUNDATION

What we are building - and what already exists

BiteBud is a smart infant-teething and wellness-tracking concept. The defensible core is a soft teether concept with pressure sensing plus a parent-facing dashboard that summarizes chewing frequency, relative intensity, duration, and changes over time. It supports awareness; it does not diagnose illness or determine why a baby is crying.

Physical concept

Soft, chew-safe product direction with embedded pressure sensing. The current hardware is strictly a tabletop rig.

Data layer

Press/chew event detection, event duration, relative intensity, usage history, calibration, and future pressure zones.

Caregiver experience

A reassuring dashboard with trends, daily summaries, and notable-change prompts in plain language.

Evidence-based current status

Area	What is complete	Next proof required
Product framing	Problem, core solution, competitor position, safe-claim language, and future cap boundary documented.	Freeze one competition-ready claim set and one target-customer definition.
Software	React dashboard, local API, live stream, persistent readings, calibration, session summaries, reset, and sensor engine tests.	Run against physical hardware; capture stable recordings and screenshots.
Firmware	Arduino Nano ESP32 firmware path and Wi-Fi/server setup documented.	Upload to the arriving board and confirm end-to-end readings.
Hardware	A201-25 sensor, breadboard, jumper wires, resistor, cable, and wiring layout prepared; Arduino was pending as of 13 Aug.	Complete 3.3 V divider, verify wiring, then characterize noise and repeatability.
Validation	Survey themes and parent/professional questions exist; one internal claim says 100% found alerts useful.	Recover sample size/method or do not use the percentage publicly. Run structured interviews.

Definition of success for this cycle

A safe bench demo that detects and summarizes repeatable pressure patterns; a credible product model or rendering; 5-10 documented discovery interviews; and one reusable evidence package that can be adapted for all three competitions.

02 / COMPETITIONS

One project, three competition entries

The slate prioritizes fit and reuse. Conrad rewards the full innovation journey; Diamond forces a concise venture case; Blue Ocean sharpens differentiation. A single evidence library - data, interviews, prototype media, market logic, and safe claims - feeds every submission.

Priority / competition	Why BiteBud fits	Key 2026-27 dates	Entry package we prepare
1 / Conrad Challenge FLAGSHIP	Purpose-driven STEM innovation with prototype, business plan, website, and video. BiteBud fits Health & Nutrition.	Official site confirms the 2026-27 season and annual Aug-Apr cadence. Exact new-cycle milestones were not published on the reviewed official pages; verify in portal. Internal targets: activation complete 30 Oct; final package 18 Dec; submit no later than 7 Jan.	Lean Canvas; innovation brief; 3-5 min demo video; public website; prototype evidence; business/impact model.
2 / Diamond Challenge VENTURE	Strong fit for a business or social innovation concept with proof of progress. Recommended track: Business Innovation, unless impact is explicitly prioritized.	16 Sep: window opens 14 Jan, 5 PM ET: submission 10 Feb: advancing teams 9 Mar: finalists 29-30 Apr: Summit	3-5 page concept narrative; 60-sec team/concept video; later 5-min pitch and <=15-slide deck.
3 / Blue Ocean PITCH	Virtual and efficient. Good place to demonstrate how BiteBud combines soothing, teething-specific sensing, and simple insights differently from monitors and traditional teethingers.	Registration open 21 Feb: pitch due 5 Apr: Top 100 22 Apr: Top 30 4 May: Top 10 13 May: winners	Mini-course; strategy canvas; noncustomer analysis; eliminate-reduce-raise-create grid; <=5-min YouTube pitch.

Competition decision rules

Rule	Decision
Reuse before adding	Every new artifact must strengthen at least two competitions or the prototype itself.
Evidence before polish	Do not spend a week on video production while repeatability, user evidence, or basic economics remain weak.
Safety before realism	No infant or in-mouth testing with the current rig. Show a safe tabletop demo and clearly separate future product design.
Registration checkpoint	By 18 September, confirm eligibility, advisor/coach, team roster, fees, disclosure/IP rules, and exact Conrad dates.
Stop condition	If the bench system is not stable by 27 September, pause multi-zone work and focus on signal conditioning plus a truthful single-sensor demo.

03 / TIMELINE

Master timeline: build first, package once, submit three ways

Dates below are internal working deadlines unless marked as organizer dates. They include buffer so technical setbacks do not immediately become submission crises.

Window	Prototype + evidence	Competition + story	Exit gate
16-23 Aug	Receive/confirm Nano; inspect wiring; upload firmware; first live stream.	Confirm team, advisor/coach, age/eligibility, and ownership of existing work. Create master evidence folder.	Live dashboard receives real A201 readings; no unsafe wiring.
24 Aug-6 Sep	Noise check; baseline; light/medium/strong trials; event count/duration/reset; log failures.	Draft one-sentence problem, customer, value proposition, and non-diagnostic claim set.	V1 detects 20 repeated press trials with documented results.
7-20 Sep	Known-weight or scale calibration; repeatability test; freeze demo script V1.	5 caregiver interviews + 2 professional interviews. 16 Sep: Diamond opens.	Go/no-go on simple divider. Evidence log includes method and limitations.
21 Sep-4 Oct	If needed, choose op-amp/Quickstart path. If stable, begin pressure-load puck and enclosure sketch.	Competitor matrix; market sizing assumptions; basic costs; register events.	Measurement path frozen; no new sensor purchases without a test reason.
5-18 Oct	Build V2 enclosure/press fixture; capture clean dashboard sessions.	Lean Canvas; innovation brief outline; Diamond narrative outline; website content map.	30 Oct internal Conrad activation target is on track.
19 Oct-8 Nov	Add zone architecture in software; add hardware sensors only if justified and available.	Finish interviews to 10 total; write insights memo; validate trust and cleaning concerns.	V2 is demoable in under 90 seconds.
9-29 Nov	Reliability sprint: 100 press cycles; reconnect/restart test; photo/video day.	Draft shared pitch deck; prototype evidence appendix; risk/safety page.	Data and media library frozen v1.
30 Nov-18 Dec	Fix only competition-critical bugs; create backup recorded demo.	Conrad package complete internally 18 Dec; Diamond narrative and 60-sec video draft.	Holiday buffer begins with submission-ready files.

03 / TIMELINE

Submission season and final-round preparation

Window	Primary work	Hard date / decision	Output
19 Dec-7 Jan	Final QA on Conrad brief, website, and 3-5 min demo. Submit before organizer deadline.	7 Jan internal target; verify official portal deadline.	Conrad submission receipt + archived copy.
8-14 Jan	Diamond narrative proofread, 60-sec intro video, links, and forms.	14 Jan, 5 PM ET organizer deadline.	Diamond submission receipt + archived copy.
15 Jan-7 Feb	Complete Blue Ocean mini-course and mandatory strategy frameworks. Improve demo based on feedback.	No major hardware redesign unless a finalist requirement demands it.	Strategy canvas, ERRC grid, noncustomer analysis, pitch script.
8-21 Feb	Record, edit, accessibility check, upload, and test public video URL.	21 Feb local time organizer deadline.	Blue Ocean submission + YouTube link.
22 Feb-9 Mar	Prepare 5-min pitch and <=15-slide deck; rehearse Q&A.;	Diamond: 10 Feb advancing teams; 9 Mar finalists.	Reusable pitch deck v2 + 30-question Q&A; bank.
10 Mar-4 Apr	If advanced, upgrade demo reliability and booth story. Otherwise publish a project retrospective and continue prototype work.	Blue Ocean Top 100 announced 5 Apr.	Final-round demo kit or post-cycle evidence report.
5-30 Apr	Competition presentations and contingency planning for overlapping travel/online requirements.	Blue Ocean Top 30: 22 Apr. Diamond Summit: 29-30 Apr.	Final pitch, tested backup video, travel/demo checklist.
1-13 May	Blue Ocean audience outreach and final announcements.	Top 10: 4 May. Winners: 13 May.	Cycle closeout, lessons learned, 2027-28 decision.
Jun-Aug 2027	Product-design research only after cycle review: safe materials, enclosure architecture, cleaning, power, and regulatory pathway.	Gate: decide whether to continue as research prototype, venture, or both.	Prototype V3 requirements and budget - not an infant-ready product.

Critical path



Schedule protection: Treat 18 December as the true finish line for the shared narrative, proof, website, and long-form video. January and February should be adaptation and QA, not invention.

04 / PROTOTYPES

Future prototype roadmap

Each version answers one question. Advancement is earned by evidence, not by adding more sensors.

Version / timing	Question answered	Build scope	Pass criteria
V0 / complete Software simulator	Can the data experience work before hardware arrives?	Mock readings; event detection; force mapping; history; calibration; session summary; future zone model.	Tests pass; dashboard shows coherent live and saved sessions.
V1 / Aug-Sep 2026 Single-sensor bench rig	Can BiteBud reliably detect relative pressure events?	A201-25 + 47 kOhm divider + Nano ESP32 + USB + flat load pad.	Stable baseline; repeatable event detection; light/medium/strong separation; documented limitations.
V1.5 / conditional Signal-quality upgrade	Is conditioning required for useful repeatability?	Rail-to-rail op-amp circuit or Tekscan Quickstart Board; optional load concentrator.	Lower noise and more consistent calibration than V1 in the same trial protocol.
V2 / Oct-Nov 2026 Competition demo	Can judges understand the interaction in 90 seconds?	Rigid press fixture or compact external enclosure; strain relief; reliable reconnect; polished dashboard; backup recording.	100-cycle bench test; clear demo; recovery after restart; no exposed claim ambiguity.
V2.5 / Nov 2026-Jan 2027 Multi-zone proof	Does location add insight worth the complexity?	Additional pressure sensors and zone-specific software only if the single-sensor system is stable.	Distinct zone labels with acceptable cross-talk; clear caregiver value. Otherwise defer.
V3 / Summer 2027+ Product-design study	What would a real product require?	Separate soft outer-form research; sealed electronics architecture; cleaning, power, sensor loading, and manufacturing review.	Engineering requirements reviewed by qualified safety/materials/regulatory experts. No infant use.
Future module / gated Optional cap	Do temperature/motion trends add enough value to justify a wearable?	Soft wearable research for skin-temperature and motion trends; crying-event detection remains secondary.	Only after teether validation, caregiver demand, privacy review, and skin-interface safety work.

Prototype test protocol - the minimum evidence set

Test	Required record
1. Baseline	2 minutes untouched; record average, range, drift.
2. Repeatability	20 trials each at light, medium, and strong reference loads.
3. Events	Verify count, duration, threshold crossing, and no double-counting.
4. Reliability	100-cycle session plus unplug/reconnect and server restart.
5. Calibration	Use known weights or a kitchen scale; report setup and uncertainty.
6. Demo	90-second live path plus an offline backup video and screenshots.

05 / EXECUTION

Operating system: weekly work, gates, and shared assets

Workstream	Weekly owner action	Shared deliverable
Hardware + firmware	Run the planned protocol; log wiring, versions, calibration, failures, and photos.	Evidence log, parts list, test tables, reproducible setup guide.
Software + data	Keep dashboard stable; prioritize real sensor reliability, exportable evidence, and demo recovery.	Tagged demo build, screenshots, sample sessions, automated test results.
User + market	Conduct scripted interviews; separate observations from assumptions; avoid leading questions.	Interview notes, insight summary, customer/beneficiary definition, trust requirements.
Competition writing	Maintain one master narrative and adapt length/format rather than rewriting facts.	Claim sheet, competition matrix, narrative, Lean Canvas, economics.
Media + pitch	Capture progress continuously; script from evidence; time every version.	60-sec intro, 3-5 min demo, <=15-slide deck, Q&A; bank.
Safety + ethics	Review every claim and demo for non-diagnostic wording, privacy, and prototype boundaries.	Safety statement, limitations, data/privacy assumptions, prohibited-demo checklist.

Weekly rhythm

MON / choose Select the one technical risk and one story risk that matter most this week.	MIDWEEK / test Run a repeatable test or interview; save raw evidence and notes.	FRI / integrate Update the dashboard, evidence log, master narrative, and risk register.	SUN / review Score the exit gate, decide next week, and back up the demo package.
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Decision gates

Gate	Date	Go if...	If not...
G1 / signal	6 Sep	Real readings are stable enough to test.	Fix wiring, firmware, or sampling before calibration.
G2 / V1	27 Sep	Levels and events repeat acceptably.	Trigger V1.5 signal-conditioning comparison.
G3 / scope	18 Oct	Single sensor tells a clear story.	Defer multi-zone work; strengthen evidence and fixture.
G4 / evidence	29 Nov	Data, interviews, costs, media, and limits are documented.	Cut nonessential features; close evidence gaps.
G5 / submit	18 Dec	Conrad package passes factual, link, timing, and safety review.	Use buffer; no speculative claims to fill gaps.

06 / NEXT ACTIONS

Immediate plan, risks, and source notes

Next 14 days

When	Action	Done when
Today	Confirm whether Conrad, Diamond, and Blue Ocean are the actual team choices; record registration status and advisor/coach.	Competition register has owners, credentials, fees, and exact dates.
Within 48 hours of Nano arrival	Complete power-off wiring check, connect board, upload firmware, and capture first live session.	Timestamped screenshot/video and setup notes saved.
Week 1	Run baseline and repeat-press tests; do not tune thresholds only by eye.	Raw trial table plus chosen threshold rationale.
Week 2	Run light/medium/strong and known-load trials; draft 90-second demo.	V1 gate decision and a truthful limitations paragraph.
By 18 Sep	Register/confirm eligibility; conduct first 5 caregiver interviews; choose Diamond track.	No unresolved eligibility or team/IP issues.

Top risks and controls

Risk	Control
Unsafe interpretation	Use relative intensity and trend language. State clearly that the rig is not for in-mouth use and BiteBud is not diagnostic.
Sensor nonlinearity / loading variability	Use a flat load concentrator, a repeatable fixture, documented calibration, and a V1.5 conditioning comparison only if needed.
Weak validation evidence	Report sample size and method. Never use the internal "100%" survey claim unless its source and denominator are recovered.
Scope creep	Cap and temperature/motion sensing remain future work. Multi-zone is conditional.
Deadline collision	Finish shared evidence by 29 Nov and shared narrative by 18 Dec; maintain backup demo media and submission receipts.
Public disclosure / IP	Review each competition's disclosure terms before posting technical details; do not publish secrets or personal caregiver data.

Source and date notes

Project sources: local BiteBud README, project context, workflow, hardware plan, future prototype plan, firmware/software structure, and Git history, reviewed 16 Aug 2026. Hardware arrival status is last documented 13 Aug 2026.

Competition sources: Conrad Challenge official site (2026-27 season confirmed; annual August-April cadence; exact new-cycle dates not available on reviewed official pages): <https://conrad.spacecenter.org/about-challenge/> . Diamond Challenge official 2027 competition page: <https://diamondchallenge.org/competition/> . Blue Ocean official 2027 competition page: <https://blueoceancompetition.org/compete/> .

Assumptions to confirm: team members remain eligible high-school students; Jack Wang and Dhruv Kumar are the active team; an adult advisor/coach is available; no competition registrations are yet confirmed; the Nano arrival status may have changed; and competition rules remain subject to organizer updates.